

# Malware Analysis Report For MachO-OSX-ppc-openssl-1.0

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## Binary Type

Mach-O (Mach Object). (Approved)

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## Analyzing File

- **File Name:** MachO-OSX-ppc-openssl-1.0
  - **File Extension:** .1h
  - **File Type:** Mach-O (Mach Object).
  - **MD5 Hash:** bae4089c1ed3d12c3529b513bb15a8b2
  - **SHA256 Hash:** 06b0515f7779a51a8362a32428d4f8de1bd30021ed2ab5092cb8f9a87a0fe4fd
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## File Identification

File detected as a valid macOS/iOS executable.

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## Mach-O File Analysis

**Mach-O File Analysis:** True

### Mach-O File Imports

- libSystem.B.dylib
- libcrypto.1.0.0.dylib
- libgcc\_s.1.dylib
- libssl.1.0.0.dylib

### Mach-O File Exports

No exported functions were found.

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## String Analysis

### ASCII Strings

- 8\_\_PAGEZERO
- \_\_TEXT
- \_\_text
- \_\_TEXT
- \_symbolstub1
- \_\_TEXT
- \_\_cstring
- \_\_TEXT
- \_\_const

- \_\_TEXT
- \_\_literal4
- \_\_TEXT
- \_\_literal8
- \_\_TEXT
- \_\_DATA
- \_\_data
- \_\_DATA
- \_\_dyld
- \_\_DATA
- \_/asymbol\_ptr
- \_\_DATA
- \_n/symbol\_ptr
- \_\_DATA
- \_\_const
- \_\_DATA
- \_\_DATA
- \_\_common
- \_\_DATA
- 8\_\_LINKEDIT
- /usr/lib/dyld
- /usr/local/Cellar/openssl/1.0.1h/lib/libssl.1.0.0.dylib
- /usr/local/Cellar/openssl/1.0.1h/lib/libcrypto.1.0.0.dylib
- /usr/lib/libgcc\_s.1.dylib
- /usr/lib/libSystem.B.dylib
- V\;iVp<\_
- 8:BVh;)V<<\_
- x:BR\$<\_
- 9"Jh8!
- V1@>H
- V1@>H
- V1@>H
- X:BJ@<\_
- Cx};Kx}YSx
- U" 6j`
- Cx}2Kx}QSx
- x}%Kx8
- Cx}2Kx}QSx|v
- ;BRh;)X
- ;"@T<\_
- 4T@F>/
- H;B5t;
- ;",,<\_
- ;b#@<\_
- H:b#H<\_
- kx8B6(
- x9)W@H
- :B0H<\_
- ;".!<\_
- .}b[xH
- /}BSx@
- 8b#T=?

- = C0<\_
- x}'KxH
- ;)gl;a
- ;i^d:a
- +p8cgdH
- :)^t=?
- A<@C0o
- 8:B^X<\_
- x}{q.@
- ~[H.=?
- @.U)@.T
- KxUk@.
- @.U)@.T
- KxUk@.
- @.U)@.T
- KxUk@.
- :!Nx??
- wX,;b}4<\_
- Cx};Kx}{Sx|y
- 3x}9Kx|z
- Cx}6Kx}USxH
- 3x}9Kx|z
- }"X.9)
- 9b`(9I
- 9b`89I
- 9b`L9I
- 9b``9I
- 9b`p9I
- Cx}i[x
- 9Bdt<\_
- ;bWl<\_
- p:bOP:)O<>\_
- U)@.|B
- :BDP~D
- :b6T:l6H>
- x9)-x8
- Cx};Kx}YSxH
- 0TB >|P
- 8TB >|P
- 
- tTB >|P
- Cx};Kx}ZSx
- ;x};Kx}YSx;
- 
- %s => %s
- list-standard-commands
- list-message-digest-commands
- list-message-digest-algorithms
- list-cipher-commands
- list-cipher-algorithms
- list-public-key-algorithms
- Name: %s

- Type: Alias to %s
- External
- Builtin
- Type: %s Algorithm
- OID: %s
- (none)
- PEM string: %s
- openssl:Error: '%s' is an invalid command.
- Standard commands
- Message Digest commands (see the `dgst' command for more details)
- Cipher commands (see the `enc' command for more details)
- OPENSSLDEBUGMEMORY
- OPENSSL\_FIPS
- FIPS mode not supported.
- OPENSSL\_CONF
- SSLEAY\_CONF
- WARNING: can't open config file: %s
- OpenSSL>
- error in %s
- bad exit
- invalid mode
- type out of bounds
- already locked
- not locked
- CRYPTOunlock on write lock
- CRYPTOunlock on read lock
- openssl (lockdbgcb): %s (mode=%d, type=%d) at %s:%d
- verify
- asn1parse
- dhparam
- passwd
- errstr
- rsautl
- dsaparam
- ecparam
- genrsa
- gendsa
- genpkey
- s\_server
- s\_client
- s\_time
- version
- crl2pkcs7
- sess\_id
- ciphers
- pkcs12
- pkeyparam
- pkeyutil
- engine
- rmd160
- aes-128-cbc
- aes-128-ecb

- aes-192-cbc
- aes-192-ecb
- aes-256-cbc
- aes-256-ecb
- camellia-128-cbc
- camellia-128-ecb
- camellia-192-cbc
- camellia-192-ecb
- camellia-256-cbc
- camellia-256-ecb
- base64
- rc4-40
- des-ecb
- des-edc
- des-edc3
- des-cbc
- des-edc-cbc
- des-edc3-cbc
- des-cfb
- des-edc-cfb
- des-edc3-cfb
- des-ofb
- des-edc-ofb
- des-edc3-ofb
- idea-cbc
- idea-ecb
- idea-cfb
- idea-ofb
- seed-cbc
- seed-ecb
- seed-cfb
- seed-ofb
- rc2-cbc
- rc2-ecb
- rc2-cfb
- rc2-ofb
- rc2-64-cbc
- rc2-40-cbc
- bf-cbc
- bf-ecb
- bf-cfb
- bf-ofb
- cast5-cbc
- cast5-ecb
- cast5-cfb
- cast5-ofb
- cast-cbc
- certificate file
- -CApath
- -CAfile
- -untrusted
- -trusted

- -CRLfile
- -engine
- -verbose
- Error loading file %s
- Error loading directory %s
- untrusted certificates
- trusted certificates
- other CRLs
- usage: verify [-verbose] [-CApath path] [-CAfile file] [-purpose purpose] [-crl\_check]
- [-atime timestamp]
- [-engine e]
- cert1 cert2 ...
- recognized usages:
- [CRL path]
- %error %d at %d depth lookup:%s
- Memory allocation failure
- -inform
- -noout
- -offset
- -length
- -dlimit
- -strparse
- -genstr
- -genconf
- unknown option %s
- %s [options]
- where options are
- -inform arg input format - one of DER PEM
- -in arg input file
- -out arg output file (output format is always DER
- -noout arg don't produce any output
- -offset arg offset into file
- -length arg length of section in file
- -i indent entries
- -dump dump unknown data in hex form
- -dlimit arg dump the first arg bytes of unknown data in hex form
- -oid file file of extra oid definitions
- -strparse offset
- a series of these can be used to 'dig' into multiple
- ASN1 blob wrappings
- -genstr str string to generate ASN1 structure from
- -genconf file file to generate ASN1 structure from
- problems opening %s
- default
- Can't find 'asn1' in '%s'
- Error on line %ld of config file '%s'
- Error loading config file '%s'
- '%s' is an invalid number
- Error parsing structure
- OBJECT
- Can't parse %s type
- Error: offset too large

- Error writing output
- string is too short, it needs to be at least %d bytes long
- string is too long, it needs to be less than %d bytes long
- parameter error "%s"
- param:
- Unknown algorithm %.\*s
- Can't open parameter file %s
- Error reading parameter file %s
- Key Type does not match parameters
- Internal error: can't find key algorithm
- Error allocating keygen context
- Error initializing keygen context
- Error setting RSA keysize
- -outform
- -keygen\_engine
- Can't find keygen engine %s
- -pubkey
- -config
- -keyform
- -keyout
- -passin
- -passout
- -newkey
- -pkeyopt
- -sigopt
- -batch
- -newhdr
- -modulus
- -verify
- -nodes
- -nameopt
- -reqopt
- -subject
- -asn1-kludge
- -no-asn1-kludge
- -multivalue-rdn
- -set\_serial
- -extensions
- -reqexts
- %s [options]
- where options are
- -inform arg input format - DER or PEM
- -outform arg output format - DER or PEM
- -in arg input file
- -out arg output file
- -text text form of request
- -pubkey output public key
- -noout do not output REQ
- -verify verify signature on REQ
- -modulus RSA modulus
- -nodes don't encrypt the output key
- -engine e use engine e, possibly a hardware device

- -subject output the request's subject
- -passin private key password source
- -key file use the private key contained in file
- -keyform arg key file format
- -keyout arg file to send the key to
- -rand file%cfile%c...
- load the file (or the files in the directory) into
- the random number generator
- -newkey rsa:bits generate a new RSA key of 'bits' in size
- -newkey dsa:file generate a new DSA key, parameters taken from CA in 'file'
- -newkey ec:file generate a new EC key, parameters taken from CA in 'file'
- -[digest] Digest to sign with (md5, sha1, md2, mdc2, md4)
- -config file request template file.
- -subj arg set or modify request subject
- -multivalue-rdn enable support for multivalued RDNs
- -new new request.
- -batch do not ask anything during request generation
- -x509 output a x509 structure instead of a cert. req.
- -days number of days a certificate generated by -x509 is valid for.
- -set\_serial serial number to use for a certificate generated by -x509.
- -newhdr output "NEW" in the header lines
- -asn1-kludge Output the 'request' in a format that is wrong but some CA's have been reported as requiring
- -extensions .. specify certificate extension section (override value in config file)
- -reqexts .. specify request extension section (override value in config file)
- -utf8 input characters are UTF8 (default ASCII)
- -nameopt arg - various certificate name options
- -reqopt arg - various request text options
- Error getting passwords
- Using configuration from %s
- error on line %ld of %s
- Unable to load config info from %s
- oid\_file
- default\_md
- x509\_extensions
- Error Loading extension section %s
- input\_password
- output\_password
- string\_mask
- Invalid global string mask setting %s
- req\_extensions
- Error Loading request extension section %s
- Private Key
- RANDFILE
- default\_bits
- private key length is too short,
- it needs to be at least %d bits, not %ld
- Generating a %ld bit %s private key
- Error Generating Key
- default\_keyfile
- writing new private key to stdout
- writing new private key to '%s'



- encryptrsakey
- encrypt\_key
- bad input format specified for X509 request
- unable to load X509 request
- you need to specify a private key
- prompt
- distinguished\_name
- unable to find '%s' in config
- unable to get '%s' section
- attributes
- error, no objects specified in config file
- You are about to be asked to enter information that will be incorporated
- into your certificate request.
- What you are about to enter is what is called a Distinguished Name or a DN.
- There are quite a few fields but you can leave some blank
- For some fields there will be a default value,
- If you enter '.', the field will be left blank.
- \_default
- \_value
- %s\_default
- Name '%s' too long
- %s\_value
- %s\_min
- %s\_max
- %s [%s]:
- weird input :-(
- Please enter the following 'extra' attributes
- to be sent with your certificate request
- Error adding attribute
- No template, please set one up.
- problems making Certificate Request
- Cannot modify certificate subject
- Modifying Request's Subject
- old subject=
- ERROR: cannot modify subject
- new subject=
- verify failure
- verify OK
- Error getting public key
- subject=
- Modulus=unavailable
- Modulus=
- Wrong Algorithm type
- bad output format specified for outfile
- unable to write X509 request
- unable to write X509 certificate
- Read Error in %s
- Verified OK
- Verification Failure
- Error Verifying Data
- Error Signing Data
- %s-%s(%s)=

- %s(%s)=
- (%s)=
- dgst.c
- out of memory
- -prverify
- -signature
- -binary
- -fips-fingerprint
- etaonrshdlcupfm
- -non-fips-allow
- -macopt
- No signature to verify: use the -signature option
- unknown option '%s'
- options are
- -c to output the digest with separating colons
- -r to output the digest in coreutils format
- -d to output debug info
- -hex output as hex dump
- -binary output in binary form
- -sign file sign digest using private key in file
- -verify file verify a signature using public key in file
- -prverify file verify a signature using private key in file
- -keyform arg key file format (PEM or ENGINE)
- -out filename output to filename rather than stdout
- -signature file signature to verify
- -sigopt nm:v signature parameter
- -hmac key create hashed MAC with key
- -mac algorithm create MAC (not necessarily HMAC)
- -macopt nm:v MAC algorithm parameters or key
- -engine e use engine e, possibly a hardware device.
- Error getting password
- (stdout)
- Error opening output file %s
- MAC and Signing key cannot both be specified
- key file
- MAC parameter error "%s"
- Error generating key
- Error getting context
- Error setting context
- Error setting digest %s
- Error opening signature file %s
- Error reading signature file %s
- %s to use the %s message digest algorithm
- -check
- -outform arg output format - one of DER PEM
- -out arg output file
- -check check the DH parameters
- -text print a text form of the DH parameters
- -C Output C code
- -noout no output
- -engine e use engine e, possibly a hardware device.
- bad input format specified

- unable to load DH parameters
- p value is not prime
- p value is not a safe prime
- unable to check the generator value
- the g value is not a generator
- DH parameters appear to be ok.
- OPENSSL\_malloc
- static unsigned char dh%d\_p[]={
- 0x%02X,
- static unsigned char dh%d\_g[]={
- DH \*get\_dh%d()
- DH \*dh;
- if ((dh=DH\_new()) == NULL) return(NULL);
- dh->p=BNbin2bn(dh%dp,sizeof(dh%d\_p),NULL);
- dh->g=BNbin2bn(dh%dg,sizeof(dh%d\_g),NULL);
- if ((dh->p == NULL) || (dh->g == NULL))
- return(NULL);
- return(dh);
- unable to write DH parameters
- -dsaparam
- %s [options] [numbits]
- -dsaparam read or generate DSA parameters, convert to DH
- -2 generate parameters using 2 as the generator value
- -5 generate parameters using 5 as the generator value
- numbits number of bits in to generate (default 512)
- - load the file (or the files in the directory) into
- the random number generator
- generator may not be chosen for DSA parameters
- warning, not much extra random data, consider using the -rand option
- %ld semi-random bytes loaded
- Generating DSA parameters, %d bit long prime
- Generating DH parameters, %d bit long safe prime, generator %d
- This is going to take a long time
- unable to load DSA parameters
- dhparam.c
- #ifndef HEADERDHH
- #include
- #endif
- static unsigned char dh%dp[]={
- static unsigned char dh%dg[]={
- { DHfree(dh); return(NULL); }
- dh->length = %ld;
- hex string is too long
- non-hex digit
- %s is an unknown cipher
- -nopad
- -nosalt
- -debug
- -base64
- -bufsize

- *-kfile*
- *unable to read key from '%s'*
- *zero length password*
- *%-14s input file*
- *-in*
- *%-14s output file*
- *-out*
- *%-14s pass phrase source*
- *-pass*
- *%-14s encrypt*
- *%-14s decrypt*
- *%-14s base64 encode/decode, depending on encryption flag*
- *-a/-base64*
- *%-14s passphrase is the next argument*
- *%-14s passphrase is the first line of the file argument*
- *%-14s the next argument is the md to use to create a key*
- *%-14s from a passphrase. One of md2, md5, sha or sha1*
- *%-14s salt in hex is the next argument*
- *%-14s key/iv in hex is the next argument*
- *-K/-iv*
- *%-14s print the iv/key (then exit if -P)*
- *%-14s buffer size*
- *-bufsize*
- *%-14s disable standard block padding*
- *%-14s use engine e, possibly a hardware device.*
- *-engine e*
- *Cipher Types*
- *AEAD ciphers not supported by the enc utility*
- *%s is an unsupported message digest type*
- *invalid 'bufsize' specified.*
- *bufsize=%d*
- *OPENSSLmalloc failure %ld*
- *encryption*
- *decryption*
- *enter %s %s password:*
- *bad password read*
- *invalid hex salt value*
- *error writing output file*
- *error reading input file*
- *bad magic number*
- *invalid hex iv value*
- *iv undefined*
- *invalid hex key value*
- *Error setting cipher %s*
- *bad decrypt*
- *bytes read :%8ld*
- *bytes written:%8ld*
- *-%-25s*
- *Salted\_*
- *passwd.c*
- *saltp != NULL*
- *saltmallocp != NULL*

- `salt_p != NULL`
- *Warning: truncating password to %u characters*
- `strlen(passwd) <= pw_maxlen`
- `strlen(magic) <= 4`
- `strlen(out_buf) <= 6 + 8`
- `salt_len <= 8`
- `output == outbuf + strlen(outbuf)`
- `strlen(outbuf) < sizeof(outbuf)`
- `hash != NULL`
- `-crypt`
- `-stdin`
- `-noverify`
- `-quiet`
- `-table`
- `-reverse`
- *Usage: passwd [options] [passwords]*
- `-crypt` standard Unix password algorithm (default)
- `-1` MD5-based password algorithm
- `-apr1` MD5-based password algorithm, Apache variant
- `-salt` string use provided salt
- `-in` file read passwords from file
- `-stdin` read passwords from stdin
- `-noverify` never verify when reading password from terminal
- `-quiet` no warnings
- `-table` format output as table
- `-reverse` switch table columns
- `in_stdin == 0`
- `in_stdin`
- *Password:*
- `passwd != NULL`
- `passwd != NULL`
- `passwdmain`
- `dopasswd`
- `md5crypt`
- *usage: gendh [args] [numbits]*
- `-out` file - output the key to 'file'
- `-2` - use 2 as the generator value
- `-5` - use 5 as the generator value
- `-engine e` - use engine e, possibly a hardware device.
- - load the file (or the files in the directory) into
- the random number generator
- `-stats`
- `%s`: bad error code
- *usage: errstr [-stats] ...*
- invalid revocation date `%s`
- missing hold instruction
- invalid object identifier `%s`
- missing compromised time
- invalid compromised time `%s`
- invalid reason code `%s`
- `PRINTABLE:`

- T61STRING:'
- IA5STRING:'
- UNIVERSALSTRING:'
- ASN.1 %2d:'
- \0x%02X
- Unknown CRL reason %s
- holdInstruction
- Invalid object identifier %s
- Invalid time format %s. Need YYYYMMDDHHMMSSZ
- CAkeyTime
- keyTime
- malloc error
- The Subject's Distinguished Name is as follows
- emailAddress type needs to be of type IA5STRING
- The string contains characters that are illegal for the ASN.1 type
- %s:unknown object type in 'policy' configuration
- optional
- supplied
- The %s field needed to be supplied and was missing
- The mandatory %s field was missing
- The %s field does not exist in the CA certificate,
- the 'policy' is misconfigured
- The %s field needed to be the same in the
- CA certificate (%s) and the request (%s)
- %s:invalid type in 'policy' configuration
- The subject name appears to be ok, checking data base for clashes
- ERROR:There is already a certificate for %s
- ERROR:Serial number %s has already been issued,
- check the database/serialfile for corruption
- *The matching entry has the following details*
- *Expired*
- *Revoked*
- *invalid type, Data base error*
- *Was revoked on:%s*
- *Expires on :%s*
- *Serial Number :%s*
- *File name :%s*
- *Subject Name :%s*
- *Everything appears to be ok, creating and signing the certificate*
- *Extra configuration file found*
- *ERROR: adding extensions in section %s*
- *Successfully added extensions from file.*
- *Successfully added extensions from config*
- *ERROR: adding extensions from request*
- *Certificate Details:*
- *Certificate is to be certified until*
- *(%ld days)*
- *Sign the certificate? [y/n]:*
- *CERTIFICATE WILL NOT BE CERTIFIED: I/O error*
- *CERTIFICATE WILL NOT BE CERTIFIED*
- *unknown*
- *failed to update database*

- *TXTDB* error number %ld
- Error reading certificate request in %s
- Check that the request matches the signature
- Certificate request and CA private key do not match
- error unpacking public key
- Signature verification problems....
- Signature did not match the certificate request
- Signature ok
- Adding Entry with serial number %s to DB for %s
- ERROR:name does not match %s
- ERROR:Already revoked, serial number %s
- Revoking Certificate %s.
- Error in revocation arguments
- *-createserial*
- *-startdate*
- *-enddate*
- *-policy*
- *-keyfile*
- *-selfsign*
- *-outdir*
- *-notext*
- *-preserveDN*
- *-noemailDN*
- *-gencrl*
- *-msiehack*
- *-crl*days
- *-crl*hours
- *-crl*sec
- *-infiles*
- *-sscert*
- *-spkac*
- *-revoke*
- *-extfile*
- *-status*
- *-updatedb*
- *-crl*exts
- *-crl*reason
- *-crl*hold
- *-crl*compromise
- *-crl*CAcompromise
- openssl.cnf
- error loading the config file '%s'
- error on line %ld of config file '%s'
- defaultca
- *variable lookup failed for %s::%s*
- *uniquesubject*
- database
- Malloc failure
- Serial %s not present in db.
- %s=Valid (%c)
- %s=Revoked (%c)
- %s=Expired (%c)

- %s=Suspended (%c)
- %s=Unknown (%c).
- Error verifying serial %s!
- privatekey
- CA private key
- certificate
- CA certificate
- CA certificate and CA private key do not match
- preserve
- msiehack
- nameopt
- Invalid name options: "%s"
- certopt
- Invalid certificate options: "%s"
- copyextensions
- Invalid extension copy option: "%s"
- newcertsdir
- there needs to be defined a directory for newcertificate to be placed in
- I am unable to access the %s directory
- %s need to be a directory
- entry %d: not revoked yet, but has a revocation date
- in entry %d
- entry %d: invalid expiry date
- entry %d: bad serial number length (%d)
- entry %d: bad serial number characters, char pos %ld, char is '%c'
- %d entries loaded from the database
- generating index
- Updating %s ...
- %s=Expired
- No entries found to mark expired
- Done. %d entries marked as expired
- ERROR: loading the config file '%s'
- ERROR: on line %ld of config file '%s'
- Successfully loaded extensions file %s
- extensions
- no default digest
- emailindn
- message digest is %s
- policy
- policy is %s
- serial
- defaultstartdate
- start date is invalid, it should be YYMMDDHHMMSSZ or YYYYMMDDHHMMSSZ
- defaultenddate
- end date is invalid, it should be YYMMDDHHMMSSZ or YYYYMMDDHHMMSSZ
- defaultdays
- cannot lookup how many days to certify for
- error while loading serial number
- next serial number is 00
- next serial number is %s
- unable to find 'section' for %s
- no name/value pairs found in %s



- unable to load Netscape SPKAC structure
- Netscape SPKAC structure not found in %s
- Check that the SPKAC request matches the signature
- error unpacking SPKAC public key
- signature verification failed on SPKAC public key
- Signature did not match the certificate
- %d out of %d certificate requests certified, commit? [y/n]
- CERTIFICATION CANCELED: I/O error
- CERTIFICATION CANCELED
- Write out database with %d new entries
- writing new certificates
- certificate file name too long
- writing %s
- Data Base Updated
- crlextensions
- *Error Loading CRL extension section %s*
- *crlnumber*
- *error while loading CRL number*
- *defaultcrl days*
- *defaultcrl hours*
- *cannot lookup howlong until the next CRL is issued*
- *making CRL*
- *error setting CRL nextUpdate*
- *signing CRL*
- *no input files*
- *unspecified*
- *keyCompromise*
- *CACompromise*
- *affiliationChanged*
- *superseded*
- *cessationOfOperation*
- *certificateHold*
- *removeFromCRL*
- *usage: ca args*
- *-verbose - Talk alot while doing things*
- *-config file - A config file*
- *-name arg - The particular CA definition to use*
- *-gencrl - Generate a newCRL*
- *-crl days days - Days is when the next CRL is due*
- *-crl hours hours - Hours is when the next CRL is due*
- *-startdate YYMMDDHHMMSSZ - certificate validity notBefore*
- *-enddate YYMMDDHHMMSSZ - certificate validity notAfter (overrides -days)*
- *-days arg - number of days to certify the certificate for*
- *-md arg - md to use, one of md2, md5, sha or sha1*
- *-policy arg - The CA 'policy' to support*
- *-keyfile arg - private key file*
- *-keyform arg - private key file format (PEM or ENGINE)*
- *-key arg - key to decode the private key if it is encrypted*
- *-cert file - The CA certificate*
- *-selfsign - sign a certificate with the key associated with it*
- *-in file - The input PEM encoded certificate request(s)*
- *-out file - Where to put the output file(s)*

- *-outdir dir* - Where to put output certificates
- *-infile ....* - The last argument, requests to process
- *-spkac file* - File contains DN and signed public key and challenge
- *-sscert file* - File contains a self signed cert to sign
- *-preserveDN* - Don't re-order the DN
- *-noemailDN* - Don't add the EMAIL field into certificate' subject
- *-batch* - Don't ask questions
- *-msiehack* - msie modifications to handle all those universal strings
- *-revoke file* - Revoke a certificate (given in file)
- *-subj arg* - Use arg instead of request's subject
- *-utf8* - input characters are UTF8 (default ASCII)
- *-multivalue-rdn* - enable support for multivalued RDNs
- *-extensions ..* - Extension section (override value in config file)
- *-extfile file* - Configuration file with X509v3 extensions to add
- *-crlxsts ..* - CRL extension section (override value in config file)
- *-engine e* - use engine e, possibly a hardware device.
- *-status serial* - Shows certificate status given the serial number
- *-updatedb* - Updates db for expired certificates
- *-print*
- *-printcerts*
- *-inform arg* input format - DER or PEM
- *-outform arg* output format - DER or PEM
- *-printcerts* print any certs or crl in the input
- *-text* print full details of certificates
- *-noout* don't output encoded data
- *bad input format specified for pkcs7 object*
- *unable to load PKCS7 object*
- *unable to write pkcs7 object*
- *-nocrl*
- *-certfile*
- *-certfile arg* certificates file of chain to a trusted CA
- *(can be used more than once)*
- *-nocrl* no crl to load, just certs from '-certfile'
- *bad input format specified for input crl*
- *unable to load CRL*
- *error opening the file, %s*
- *error reading the file, %s*
- *error loading certificates*
- *-hashold*
- *-issuer*
- *-lastupdate*
- *-nextupdate*
- *-fingerprint*
- *-crlnumber*
- *Error initialising X509 store*
- *Error getting CRL issuer certificate*
- *Error getting CRL issuer public key*
- *issuer=*
- *crlNumber=*
- 
- *lastUpdate=*
- *nextUpdate=*

- %s Fingerprint=
- %02X%c
- unable to write CRL
- usage: `crl args`
- `-inform arg` - input format - default PEM (DER or PEM)
- `-outform arg` - output format - default PEM
- `-text` - print out a text format version
- `-in arg` - input file - default stdin
- `-out arg` - output file - default stdout
- `-hash` - print hash value
- `-hashold` - *print old-style (MD5) hash value*
- `-fingerprint` - *print the crl fingerprint*
- `-issuer` - *print issuer DN*
- `-lastupdate` - *lastUpdate field*
- `-nextupdate` - *nextUpdate field*
- `-crlnumber` - *print CRL number*
- `-noout` - *no CRL output*
- `-CAfile name` - *verify CRL using certificates in file "name"*
- `-CApath dir` - *verify CRL using certificates in "dir"*
- `-sgckey`
- `-pubin`
- `-pubout`
- `-RSAPublicKeyin`
- `-RSAPublicKeyout`
- `-pvk-strong`
- `-pvk-weak`
- `-pvk-none`
- `-inform arg` *input format - one of DER NET PEM*
- `-outform arg` *output format - one of DER NET PEM*
- `-in arg` *input file*
- `-sgckey` *Use IIS SGC key format*
- `-passin arg` *input file pass phrase source*
- `-out arg` *output file*
- `-passout arg` *output file pass phrase source*
- `-des` *encrypt PEM output with cbc des*
- `-des3` *encrypt PEM output with ede cbc des using 168 bit key*
- `-idea` *encrypt PEM output with cbc idea*
- `-seed` *encrypt PEM output with cbc seed*
- `-aes128, -aes192, -aes256`
- *encrypt PEM output with cbc aes*
- `-camellia128, -camellia192, -camellia256`
- *encrypt PEM output with cbc camellia*
- `-text` *print the key in text*
- `-noout` *don't print key out*
- `-modulus` *print the RSA key modulus*
- `-check` *verify key consistency*
- `-pubin` *expect a public key in input file*
- `-pubout` *output a public key*
- `-engine e` *use engine e, possibly a hardware device.*
- *Only private keys can be checked*
- *Public Key*
- *RSA key ok*

- *RSA key error: %s*
- *writing RSA key*
- *unable to write key*
- *-inkey*
- *-certin*
- *-asn1parse*
- *-hexdump*
- *-encrypt*
- *-decrypt*
- *Usage: rsautl [options]*
- *-in file input file*
- *-out file output file*
- *-inkey file input key*
- *-keyform arg private key format - default PEM*
- *-pubin input is an RSA public*
- *-certin input is a certificate carrying an RSA public key*
- *-ssl use SSL v2 padding*
- *-rawuse no padding*
- *-pkcs use PKCS#1 v1.5 padding (default)*
- *-oaep use PKCS#1 OAEP*
- *-sign sign with private key*
- *-verify verify with public key*
- *-encrypt encrypt with public key*
- *-decrypt decrypt with private key*
- *-hexdump hex dump output*
- *-passin arg pass phrase source*
- *A private key is needed for this operation*
- *Certificate*
- *Error getting RSA key*
- *Error Reading Input File*
- *Error Reading Output File*
- *rsautl.c*
- *Error reading input Data*
- *RSA operation error*
- *-inform arg input format - DER or PEM*
- *-outform arg output format - DER or PEM*
- *-modulus print the DSA public value*
- *read DSA key*
- *unable to load Key*
- *Public Key=*
- *writing DSA key*
- *unable to write private key*
- *-genkey*
- *%s [options] [bits]*
- *-text print as text*
- *-genkey generate a DSA key*
- *-rand files to use for random number input*
- *number number of bits to use for generating private key*
- *dsaparam.c*
- *needrand*
- *Error allocating DSA object*
- *This could take some time*

- Error, DSA key generation failed
- `static unsigned char dsa%dp[]={`
- `static unsigned char dsa%dq[]={`
- `static unsigned char dsa%dg[]={`
- `DSA *getdsa%d()`
- `DSA dsa;`
- `if ((dsa=DSA_new()) == NULL) return(NULL);`
- `dsa->p=BNbin2bn(dsa%dp,sizeof(dsa%d_p),NULL);`
- `dsa->q=BNbin2bn(dsa%dq,sizeof(dsa%d_q),NULL);`
- `dsa->g=BNbin2bn(dsa%dg,sizeof(dsa%d_g),NULL);`
- `if ((dsa->p == NULL) || (dsa->q == NULL) || (dsa->g == NULL))`
- `{ DSA_free(dsa); return(NULL); }`
- `return(dsa);`
- *unable to write DSA parameters*
- *dsaparam\_main*
- *-conv\_form*
- *compressed*
- *uncompressed*
- *hybrid*
- *-param\_enc*
- *named\_curve*
- *explicit*
- *-param\_out*
- *-des encrypt PEM output, instead of 'des' every other*
- *cipher supported by OpenSSL can be used*
- *-text print the key*
- *-param\_out print the elliptic curve parameters*
- *-conv\_form arg specifies the point conversion form*
- *possible values: compressed*
- *uncompressed (default)*
- *hybrid*
- *-param\_enc arg specifies the way the ec parameters are encoded*
- *in the asn1 der encoding*
- *possible values: named\_curve (default)*
- *explicit*
- *read EC key*
- *bad input format specified for key*
- *writing EC key*
- `static unsigned char %s_%d[] = {`
- `0x%02X`
- *-list\_curves*
- *-no\_seed*
- *-inform arg input format - default PEM (DER or PEM)*
- *-outform arg output format - default PEM*
- *-in arg input file - default stdin*
- *-out arg output file - default stdout*
- *-noout do not print the ec parameter*
- *-text print the ec parameters in text form*
- *-check validate the ec parameters*
- *-C print a 'C' function creating the parameters*
- *-name arg use the ec parameters with 'short name' name*
- *-list\_curves prints a list of all currently available curve 'short names'*

- `-conv_form` arg specifies the point conversion form
- possible values: `compressed`
- `uncompressed` (default)
- `hybrid`
- `-param_enc` arg specifies the way the ec parameters are encoded
- in the `asn1 der` encoding
- possible values: `named_curve` (default)
- `explicit`
- `-no_seed` if 'explicit' parameters are chosen do not use the seed
- `-genkey` generate ec key
- `-rand file` files to use for random number input
- `-engine e` use engine e, possibly a hardware device
- `ecparam.c`
- CURVE DESCRIPTION NOT AVAILABLE
- `%-10s`:
- `secp192r1`
- using curve name `prime192v1` instead of `secp192r1`
- `secp256r1`
- using curve name `prime256v1` instead of `secp256r1`
- unknown curve name (`%s`)
- unable to create curve (`%s`)
- unable to load elliptic curve parameters
- checking elliptic curve parameters:
- failed
- `ec_gen`
- `ec_order`
- `ec_cofactor`
- `ECGROUP *getecgroup%d(void)`
- `int ok=0;`
- `EC_GROUP *group = NULL;`
- `EC_POINT *point = NULL;`
- `BIGNUM *tmp1 = NULL, *tmp2 = NULL, *tmp_3 = NULL;`
- `if ((tmp1 = BNbin2bn(ecp%d, sizeof(ecp%d), NULL)) == NULL)`
- `goto err;`
- `if ((tmp2 = BNbin2bn(eca%d, sizeof(eca%d), NULL)) == NULL)`
- `goto err;`
- `if ((tmp3 = BNbin2bn(ecb%d, sizeof(ecb%d), NULL)) == NULL)`
- `goto err;`
- `if ((group = ECGROUPnewcurveGFp(tmp1, tmp2, tmp_3, NULL)) == NULL)`
- `goto err;`
- `/build generator /`
- `if ((tmp1 = BNbin2bn(ecgen%d, sizeof(ecgen%d), tmp_1)) == NULL)`
- `goto err;`
- `point = ECPOINTbn2point(group, tmp_1, NULL, NULL);`
- `if (point == NULL)`
- `goto err;`
- `if ((tmp2 = BNbin2bn(ecorder%d, sizeof(ecorder%d), tmp_2)) == NULL)`
- `goto err;`
- `if ((tmp3 = BNbin2bn(eccofactor%d, sizeof(eccofactor%d), tmp_3)) == NULL)`
- `goto err;`
- `if (!ECGROUPsetgenerator(group, point, tmp2, tmp_3))`
- `goto err;`

- *if (tmp\_1)*
- *BNfree(tmp1);*
- *if (tmp\_2)*
- *BNfree(tmp2);*
- *if (tmp\_3)*
- *BNfree(tmp3);*
- *if (point)*
- *ECPOINTfree(point);*
- *if (!ok)*
- *ECGROUPEfree(group);*
- *group = NULL;*
- *return(group);*
- *unable to write elliptic curve parameters*
- *ecparam\_main*
- *-CAform*
- *-CAkeyform*
- *bad number of days*
- *-signkey*
- *-CAkey*
- *-CAserial*
- *-addtrust*
- *Invalid trust object value %s*
- *-addreject*
- *Invalid reject object value %s*
- *-setalias*
- *-certopt*
- *-email*
- *-ocsp\_uri*
- *-serial*
- *-next\_serial*
- *-x509toreq*
- *-subject\_hash*
- *-subjecthashold*
- *-issuer\_hash*
- *-issuerhashold*
- *-dates*
- *-purpose*
- *-checkend*
- *-trustout*
- *-clrtrust*
- *-clrreject*
- *-alias*
- *-CAcreateserial*
- *-clrext*
- *-crltext*
- *use -clrext instead of -crltext*
- *-ocspid*
- *need to specify a CAkey if using the CA command*
- *We need a private key to sign with*
- *The certificate request appears to be corrupted*
- *It does not contain a public key*
- *Signature verification error*

- *CA Certificate*
- *2.99999.3*
- *SET.ex3*
- *SET x509v3 extension 3*
- *issuer=*
- *subject=*
- *serial=*
- 
- *Certificate purposes:*
- *%s%s :*
- *Yes (WARNING code=%d)*
- */subject:%s /*
- */issuer :%s \*/*
- *x509.c*
- *unsigned char XXXsubjectname[%d]={*
- *unsigned char XXXpublickey[%d]={*
- *unsigned char XXXcertificate[%d]={*
- *notBefore=*
- *notAfter=*
- *Getting Private key*
- *Private key*
- *Getting CA Private Key*
- *CA Private Key*
- *out of mem*
- *addword failure*
- *Getting request Private Key*
- *no request key file specified*
- *request key*
- *Generating certificate request*
- *Certificate will expire*
- *Certificate will not expire*
- *unable to write certificate*
- *error with certificate to be certified - should be self signed*
- *error with certificate - error %d at depth %d*
- *usage: x509 args*
- *-inform arg - input format - default PEM (one of DER, NET or PEM)*
- *-outform arg - output format - default PEM (one of DER, NET or PEM)*
- *-keyform arg - private key format - default PEM*
- *-CAform arg - CA format - default PEM*
- *-CAkeyform arg - CA key format - default PEM*
- *-passin arg - private key password source*
- *-serial - print serial number value*
- *-subjecthash - print subject hash value*
- *-subjecthashold - print old-style (MD5) subject hash value*
- *-issuerhash - print issuer hash value*
- *-issuerhashold - print old-style (MD5) issuer hash value*
- *-hash - synonym for -subjecthash*
- *-subject - print subject DN*
- *-email - print email address(es)*
- *-startdate - notBefore field*
- *-enddate - notAfter field*
- *-purpose - print out certificate purposes*



- *-dates - both Before and After dates*
- *-modulus - print the RSA key modulus*
- *-pubkey - output the public key*
- *-fingerprint - print the certificate fingerprint*
- *-alias - output certificate alias*
- *-noout - no certificate output*
- *-ocspid - print OCSP hash values for the subject name and public key*
- *-ocspuri - print OCSP Responder URL(s)*
- *-trustout - output a "trusted" certificate*
- *-clrtrust - clear all trusted purposes*
- *-clrreject - clear all rejected purposes*
- *-addtrust arg - trust certificate for a given purpose*
- *-addreject arg - reject certificate for a given purpose*
- *-setalias arg - set certificate alias*
- *-days arg - How long till expiry of a signed certificate - def 30 days*
- *-checkend arg - check whether the cert expires in the next arg seconds*
- *exit 1 if so, 0 if not*
- *-signkey arg - self sign cert with arg*
- *-x509toreq - output a certification request object*
- *-req - input is a certificate request, sign and output.*
- *-CA arg - set the CA certificate, must be PEM format.*
- *-CAkey arg - set the CA key, must be PEM format*
- *missing, it is assumed to be in the CA file.*
- *-CAcreateserial - create serial number file if it does not exist*
- *-CAserial arg - serial file*
- *-setserial - serial number to use*
- *-text - print the certificate in text form*
- *-C - print out C code forms*
- *-md2/-md5/-sha1/-mdc2 - digest to use*
- *-extfile - configuration file with X509V3 extensions to add*
- *-extensions - section from config file with X509V3 extensions to add*
- *-clrext - delete extensions before signing and input certificate*
- *-certopt arg - various certificate text options*
- *x509main*
- *unable to create BIO for output*
- *-aes128*
- *-aes192*
- *-aes256*
- *-camellia128*
- *-camellia192*
- *-camellia256*
- *usage: genrsa [args] [numbits]*
- *-des encrypt the generated key with DES in cbc mode*
- *-des3 encrypt the generated key with DES in ede cbc mode (168 bit key)*
- *-idea encrypt the generated key with IDEA in cbc mode*
- *-seed*
- *encrypt PEM output with cbc seed*
- *-out file output the key to 'file*
- *-f4 use F4 (0x10001) for the E value*
- *-3 use 3 for the E value*
- *load the file (or the files in the directory) into*
- *the random number generator*

- Generating RSA private key, %d bit long modulus
- e is %ld (0x%IX)
- usage: gendsa [args] dsaparam-file
- -out file - output the key to 'file'
- -des - encrypt the generated key with DES in cbc mode
- -des3 - encrypt the generated key with DES in ede cbc mode (168 bit key)
- -idea - encrypt the generated key with IDEA in cbc mode
- dsaparam-file
- - a DSA parameter file as generated by the dsaparam command
- unable to load DSA parameter file
- Generating DSA key, %d bits
- Algorithm already set!
- Algorithm %s not found
- Error initializing %s context
- -paramfile
- Parameters already set!
- Error initializing context
- -algorithm
- No keytype specified
- parameter setting error
- -genparam
- Unknown cipher %s
- Usage: genpkey [options]
- where options may be
- -out file output file
- -outform X output format (DER or PEM)
- -pass arg output file pass phrase source
- - use cipher to encrypt the key
- -engine e use engine e, possibly a hardware device.
- -paramfile file parameters file
- -algorithm alg the public key algorithm
- -pkeyopt opt:value set the public key algorithm option
- to value
- -genparam generate parameters, not key
- -text print the in text
- NB: options order may be important! See the manual page.
- Can't open output file %s
- Error generating parameters
- Bad format specified for key
- Error writing key
- Error printing key
- %4ld items in the session cache
- %4ld client connects (SSLconnect())
- %4ld client renegotiates (SSLconnect())
- %4ld client connects that finished
- %4ld server accepts (SSLaccept())
- %4ld server renegotiates (SSLaccept())
- %4ld server accepts that finished
- %4ld session cache hits
- %4ld session cache misses
- %4ld session cache timeouts

- %4ld callback cache hits
- %4ld cache full overflows (%ld allowed)
- server.pem
- server2.pem
- -accept
- verify depth is %d
- -Verify
- verify depth is %d, must return a certificate
- -context
- -certform
- -dhparam
- -namedcurve
- -dcertform
- -dcert
- -dkeyform
- -dpass
- -nocert
- -nocache
- -verifyreturnerror
- -serverpref
- -legacyrenegotiation
- -cipher
- -nbiotest
- -tlsextdebug
- -statusverbose
- -statustimeout
- -statusurl
- *Error parsing URL*
- -state
- -notmprsa
- -nodhe
- -noecdhe
- -pskhint
- -srpfile
- -srpuserseed
- -noss12
- -noss13
- -notls1
- -notls11
- -notls12
- -nocomp
- -noticket
- -tls11
- -tls12
- -dtls1
- -timeout
- -chain
- -idprefix
- -servername
- -servernamefatal
- -cert2

- *-nextprotoneg*
- *-usesrtp*
- *-keymatexport*
- *-keymatexportlen*
- *usage: sserver [args ...]*
- *-accept arg - port to accept on (default is %d)*
- *-context arg - set session ID context*
- *-verify arg - turn on peer certificate verification*
- *-Verify arg - turn on peer certificate verification, must have a cert.*
- *-cert arg - certificate file to use*
- *(default is %s)*
- *-crlcheck - check the peer certificate has not been revoked by its CA.*
- *The CRL(s) are appended to the certificate file*
- *-crlcheckall - check the peer certificate has not been revoked by its CA*
- *or any other CRL in the CA chain. CRL(s) are appened to the*
- *the certificate file.*
- *-certform arg - certificate format (PEM or DER) PEM default*
- *-key arg - Private Key file to use, in cert file if*
- *not specified (default is %s)*
- *-keyform arg - key format (PEM, DER or ENGINE) PEM default*
- *-pass arg - private key file pass phrase source*
- *-dcert arg - second certificate file to use (usually for DSA)*
- *-dcertform x - second certificate format (PEM or DER) PEM default*
- *-dkey arg - second private key file to use (usually for DSA)*
- *-dkeyform arg - second key format (PEM, DER or ENGINE) PEM default*
- *-dpass arg - second private key file pass phrase source*
- *-dhparam arg - DH parameter file to use, in cert file if not specified*
- *or a default set of parameters is used*
- *-namedcurve arg - Elliptic curve name to use for ephemeral ECDH keys.*
- *Use "openssl ecparam -listcurves" for all names*
- *(default is nistp256).*
- *-nbio - Run with non-blocking IO*
- *-nbio test - test with the non-blocking test bio*
- *-crlf - convert LF from terminal into CRLF*
- *-debug - Print more output*
- *-msg - Showprotocol messages*
- *-state - Print the SSL states*
- *-CApath arg - PEM format directory of CA's*
- *-CAfile arg - PEM format file of CA's*
- *-nocert - Don't use any certificates (Anon-DH)*
- *-cipher arg - play with 'openssl ciphers' to see what goes here*
- *-serverpref - Use server's cipher preferences*
- *-quiet - No server output*
- *-notmprsa - Do not generate a tmp RSA key*
- *-pskhint arg - PSK identity hint to use*
- *-psk arg - PSK in hex (without 0x)*
- *-srpfile file - The verifier file for SRP*
- *-srpuserseed string - A seed string for a default user salt.*
- *-ssl2 - Just talk SSLv2*
- *-ssl3 - Just talk SSLv3*
- *-tls12 - Just talk TLSv1.2*
- *-tls11 - Just talk TLSv1.1*

- -tls1 - Just talk TLSv1
- -dtls1 - Just talk DTLSv1
- -timeout - Enable timeouts
- -mtu - Set link layer MTU
- -chain - Read a certificate chain
- -noss12 - *Just disable SSLv2*
- -noss13 - Just disable SSLv3
- -notls1 - *Just disable TLSv1*
- -notls11 - *Just disable TLSv1.1*
- -notls12 - *Just disable TLSv1.2*
- -nodhe - Disable ephemeral DH
- -noecdhe - *Disable ephemeral ECDH*
- -bugs - *Turn on SSL bug compatibility*
- -www - *Respond to a 'GET /' with a status page*
- -WWW - *Respond to a 'GET / HTTP/1.0' with file ./*
- -HTTP - *Respond to a 'GET / HTTP/1.0' with file ./*
- *with the assumption it contains a complete HTTP response.*
- -engine id - *Initialise and use the specified engine*
- -idprefix arg - Generate SSL/TLS session IDs prefixed by 'arg'
- -servername host - servername for HostName TLS extension
- -servernamefatal - *on mismatch send fatal alert (default warning alert)*
- -cert2 arg - *certificate file to use for servername*
- -key2 arg - *Private Key file to use for servername, in cert file if*
- -tlsextdebug - *hex dump of all TLS extensions received*
- -noticket - disable use of RFC4507bis session tickets
- -legacyrenegotiation - *enable use of legacy renegotiation (dangerous)*
- -nextprotoneg arg - *set the advertised protocols for the NPN extension (comma-separated list)*
- -usesrtp profiles - Offer SRTP key management with a colon-separated profile list
- -keymatexport label - Export keying material using label
- -keymatexportlen len - Export len bytes of keying material (default 20)
- server certificate private key file
- server certificate file
- second server certificate private key file
- second server certificate file
- second certificate private key file
- warning: idprefix is too long, only one new session will be possible
- *warning: idprefix is too long if you use SSLv2*
- error setting 'idprefix'
- idprefix '%s' set.
- Setting secondary ctx parameters
- Setting temp DH parameters
- Using default temp DH parameters
- Using default temp ECDH parameters
- unable to create curve (nistp256)
- PSK key given or JPAKE in use, setting server callback
- error setting PSK identity hint to context
- error setting cipher list
- Cannot initialize SRP verifier file "%s":ret=%d
- ACCEPT
- Not a hex number '%s'
- Setting temp ECDH parameters

- *certstatus: callback called*
- *certstatus: can't parse AIA URL*
- *certstatus: AIA URL: %s*
- *certstatus: no AIA and no default responder URL*
- *certstatus: Can't retrieve issuer certificate.*
- *certstatus: error querying responder*
- *certstatus: ocsp response sent:*
- *sserver.c*
- *turning on non blocking io*
- *LOOKUP during accept %s*
- *LOOKUP done %s*
- *LOOKUP not successful*
- *read R BLOCK*
- *GET /stats*
- *HTTP/1.0 200 ok*
- *Content-type: text/html*
- 
- *Secure Renegotiation IS%s supported*
- *Ciphers supported in sserver binary*
- *%-11s:%-25s*
- *Ciphers common between both SSL end points:*
- *Reused,*
- *%s, Cipher is %s*
- *Client certificate*
- *no client certificate available*
- 
- *'%s' is an invalid file name*
- *'%s' contains '..' reference*
- *'%s' is an invalid path*
- *'%s' is a directory*
- *Error opening '%s'*
- *FILE:%s*
- *HTTP/1.0 200 ok*
- *Content-type: text/plain*
- *write W BLOCK*
- *SRP username = "%s"*
- *User %s doesn't exist*
- *SRP parameters set: username = "%s" info="%s"*
- *Hostname in TLS extension: "%s"*
- *Switching server context.*
- *pskservercb*
- *Error: client did not send PSK identity*
- *identitylen=%d identity=%s*
- *PSK error: client identity not found (got '%s' expected '%s')*
- *PSK client identity found*
- *Could not convert PSK key '%s' to BIGNUM*
- *psk buffer of callback is too small (%d) for key (%d)*
- *fetch PSK len=%d*
- *Error in PSK server callback*
- *Allocation error in generating RSA key*
- *Generating temp (%d bit) RSA key...*
- *CONNECTION CLOSED*

- TIMEOUT occurred
- `lfnun == 0`
- *shutdown accept socket*
- *HEARTBEATING*
- *SSLdohandshake -> %d*
- *LOOKUP renegotio during write*
- *Write BLOCK*
- *verify error:%s*
- *subject=%s*
- *issuer=%s*
- *Shared ciphers:%s*
- *(NONE)*
- *CIPHER is %s*
- *NEXTPROTO is*
- *SRTP Extension negotiated, profile=%s*
- *Reused session-id*
- *Peer has incorrect TLSv1 block padding*
- *Keying material exporter:*
- *Label: '%s'*
- *Length: %i bytes*
- *Error*
- *Keying material:*
- *LOOKUP renegotio during read*
- *Read BLOCK*
- *shutting down SSL*
- *Lets print some clear text*
- *svbody*
- 
- *Client\_identity*
- Certificate chain
- *%2d s:%s*
- *i:%s*
- Server certificate
- no peer certificate available
- Acceptable client certificate CA names
- No client certificate CA names sent
- Ciphers common between both SSL endpoints:
- SSL handshake has read %ld bytes and written %ld bytes
- Server public key is %d bit
- Compression: %s
- Expansion: %s
- Next protocol: (%d)
- *s\_client.c*
- OCSP response:
- no response sent
- response parse error
- =====
- =====
- SRP parameters:
- SRP param N and g are not known params, going to check deeper.
- SRP param N and g rejected.

- SRP user
- Can't read Password
- Can't use SSL`getservername`
- Protocols advertised by server:
- `pskclientcb`
- NULL received PSK identity hint, continuing anyway
- Received PSK identity hint '%s'
- created identity '%s' len=%d
- created PSK len=%d
- Error in PSK client callback
- localhost
- -connect
- -sess\_out
- -sess\_in
- -prexit
- -ign\_eof
- -noignoreof
- -pause
- -showcerts
- -psk\_identity
- -srpuser
- -srppass
- -srp\_strength
- SRP minimal length for N is %d
- -srp\_lateuser
- -srp\_moregroups
- -reconnect
- -legacyserverconnect
- -nolegacyserver\_connect
- -starttls
- -sslclientengine
- usage: s\_client args
- -host host - use -connect instead
- -port port - use -connect instead
- -connect host:port - who to connect to (default is %s:%s)
- -cert arg - certificate file to use, PEM format assumed
- -key arg - Private key file to use, in cert file if
- not specified but cert file is.
- -keyform arg - key format (PEM or DER) PEM default
- -reconnect - Drop and re-make the connection with the same Session-ID
- -pause - sleep(1) after each read(2) and write(2) system call
- -showcerts - show all certificates in the chain
- -debug - extra output
- -nbio\_test - more ssl protocol testing
- -state - print the 'ssl' states
- -quiet - no s\_client output
- -ign\_eof - ignore input eof (default when -quiet)
- -noignoreof - don't ignore input eof
- -psk\_identity arg - PSK identity
- -srpuser user - SRP authentication for 'user'
- -srppass arg - password for 'user'
- -srp\_lateuser - SRP username into second ClientHello message



- -srp\_moregroups - Tolerate other than the known g N values.
- -srp\_strength int - minimal length in bits for N (default %d).
- -ssl2 - just use SSLv2
- -ssl3 - just use SSLv3
- -tls1\_2 - just use TLSv1.2
- -tls1\_1 - just use TLSv1.1
- -tls1 - just use TLSv1
- -dtls1 - just use DTLSv1
- -mtu - set the link layer MTU
- -notls12/-notls11/-notls1/-noSSL3/-no\_ssl2 - turn off that protocol
- -bugs - Switch on all SSL implementation bug workarounds
- -serverpref - Use server's cipher preferences (only SSLv2)
- -cipher - preferred cipher to use, use the 'openssl ciphers' command to see what is available
- -starttls prot - use the STARTTLS command before starting TLS for those protocols that support it, where 'prot' defines which one to assume. Currently, only "smtp", "pop3", "imap", "ftp" and "xmpp" are supported.
- -sess\_out arg - file to write SSL session to
- -sess\_in arg - file to read SSL session from
- -servername host - Set TLS extension servername in ClientHello
- -tlsextdebug - hex dump of all TLS extensions received
- -status - request certificate status from server
- -no\_ticket - disable use of RFC4507bis session tickets
- -nextprotoneg arg - enable NPN extension, considering named protocols supported (comma-separated list)
- Error parsing -nextprotoneg argument
- Error getting client auth engine
- client certificate private key file
- client certificate file
- Error setting client auth engine
- PSK key given or JPAKE in use, setting client callback
- Unable to set SRP username
- Can't open session file %s
- Unable to set TLS servername extension.
- connect:errno=%d
- CONNECTED(%08X)
- getsockname:errno=%d
- EHLO openssl.client.net
- STARTTLS
- didn't found starttls in server response, try anyway...
- STARTTLS
- . CAPABILITY
- didn't found STARTTLS in server response, try anyway...
- . STARTTLS
- AUTH TLS
- 
- /stream:features>
- 
- 
- 
- Error writing session file %s

- drop connection and then reconnect
- bad select %d
- write W BLOCK
- write R BLOCK
- write X BLOCK
- shutdown
- write:errno=%d
- read W BLOCK
- read X BLOCK
- read:errno=%d
- closed
- RENEGOTIATING
- *sclient*main
- - +R:%d:%s:%f
- %d %s's in %.2fs
- +DTP:%d:%s:%s:%d
- Doing %d bit %s %s's for %ds:
- +DT:%s:%d:%d
- Doing %s for %ds on %d size blocks:
- speed.c
- -elapsed
- no EVP given
- %s is an unknown cipher or digest
- no engine given
- -multi
- no multi count given
- bad multi count
- sha256
- sha512
- whirlpool
- ripemd
- ripemd160
- aes-128-ige
- aes-192-ige
- aes-256-ige
- openssl
- dsa512
- dsa1024
- dsa2048
- rsa512
- rsa1024
- rsa2048
- rsa4096
- blowfish
- camellia
- ecdsap160
- ecdsap192
- ecdsap224
- ecdsap256
- ecdsap384
- ecdsap521

- ecdsak163
- ecdsak233
- ecdsak283
- ecdsak409
- ecdsak571
- ecdsab163
- ecdsab233
- ecdsab283
- ecdsab409
- ecdsab571
- ecdhp160
- ecdhp192
- ecdhp224
- ecdhp256
- ecdhp384
- ecdhp521
- ecdhk163
- ecdhk233
- ecdhk283
- ecdhk409
- ecdhk571
- ecdhb163
- ecdhb233
- ecdhb283
- ecdhb409
- ecdhb571
- Error: bad option or value
- Available values:
- mdc2
- md4
- md5
- hmac
- sha1
- sha256
- sha512
- idea-cbc
- seed-cbc
- rc2-cbc
- des-cbc des-ede3
- aes-128-cbc aes-192-cbc aes-256-cbc
- aes-128-ige aes-192-ige aes-256-ige
- camellia-128-cbc camellia-192-cbc camellia-256-cbc
- rsa512 rsa1024 rsa2048 rsa4096
- dsa512 dsa1024 dsa2048
- ecdsap160 ecdsap192 ecdsap224 ecdsap256 ecdsap384 ecdsap521
- ecdsak163 ecdsak233 ecdsak283 ecdsak409 ecdsak571
- ecdsab163 ecdsab233 ecdsab283 ecdsab409 ecdsab571
- ecdhp160 ecdhp192 ecdhp224 ecdhp256 ecdhp384 ecdhp521
- ecdhk163 ecdhk233 ecdhk283 ecdhk409 ecdhk571
- ecdhb163 ecdhb233 ecdhb283 ecdhb409 ecdhb571
- idea
- seed

- rc2
- des
- aes
- camellia
- rsa
- Available options:
- -evp e use EVP e.
- -decrypt time decryption instead of encryption (only EVP).
- -mr produce machine readable output.
- -multi n run n benchmarks in parallel.
- pipe failure
- Forked child %d
- dup failed
- Don't understand line '%s' from child %d
- Got: %s from %d
- Unknown type '%s' from child %d
- You have chosen to measure elapsed time instead of user CPU time.
- internal error loading RSA key number %d
- This is a key...
- 0123456789ab
- RSA sign failure. No RSA sign will be done.
- private
- RSA sign failure
- +R1:%ld:%d:%.2f
- %ld %d bit private RSA's in %.2fs
- RSA verify failure. No RSA verify will be done.
- public
- RSA verify failure
- +R2:%ld:%d:%.2f
- %ld %d bit public RSA's in %.2fs
- DSA sign failure. No DSA sign will be done.
- DSA sign failure
- +R3:%ld:%d:%.2f
- %ld %d bit DSA signs in %.2fs
- DSA verify failure. No DSA verify will be done.
- DSA verify failure
- +R4:%ld:%d:%.2f
- %ld %d bit DSA verify in %.2fs
- ECDSA failure.
- ECDSA sign failure. No ECDSA sign will be done.
- ECDSA sign failure
- +R5:%ld:%d:%.2f
- %ld %d bit ECDSA signs in %.2fs
- ECDSA verify failure. No ECDSA verify will be done.
- ECDSA verify failure
- +R6:%ld:%d:%.2f
- %ld %d bit ECDSA verify in %.2fs
- ECDH failure.
- ECDH key generation failure.
- ECDH computations don't match.
- +R7:%ld:%d:%.2f
- %ld %d-bit ECDH ops in %.2fs

- options:
- The 'numbers' are in 1000s of bytes per second processed.
- type
- %7d bytes
- +F:%d:%s
- %11.2fk
- %11.2f
- %18ssign verify sign/s verify/s
- +F2:%u:%u:%f:%f
- rsa %4u bits %8.6fs %8.6fs %8.1f %8.1f
- +F3:%u:%u:%f:%f
- dsa %4u bits %8.6fs %8.6fs %8.1f %8.1f
- %30ssign verify sign/s verify/s
- +F4:%u:%u:%f:%f
- %4u bit ecDSA (%s) %8.4fs %8.4fs %8.1f %8.1f
- %30sop op/s
- +F5:%u:%u:%f:%f
- %4u bit ecDH (%s) %8.4fs %8.1f
- hmac(md5)
- des cbc
- des ede3
- idea cbc
- seed cbc
- rc2 cbc
- rc5-32/12 cbc
- blowfish cbc
- cast cbc
- aes-128 cbc
- aes-192 cbc
- aes-256 cbc
- camellia-128 cbc
- camellia-192 cbc
- camellia-256 cbc
- aes-128 ige
- aes-192 ige
- aes-256 ige
- string to make the random number generator think it has entropy
- secp160r1
- nistp192
- nistp224
- nistp256
- nistp384
- nistp521
- nistk163
- nistk233
- nistk283
- nistk409
- nistk571
- nistb163
- nistb233
- nistb283
- nistb409

- nistb571
- localhost:4433
- -reuse
- -www option too long
- usage: s\_time
- -connect host:port - host:port to connect to (default is %s)
- -nbio - Run with non-blocking IO
- -ssl2 - Just use SSLv2
- -ssl3 - Just use SSLv3
- -bugs - Turn on SSL bug compatibility
- -new - Just time new connections
- -reuse - Just time connection reuse
- -www page - Retrieve 'page' from the site
- SSL\_CIPHER
- No CIPHER specified
- Collecting connection statistics for %d seconds
- GET %s HTTP/1.0
- %d connections in %.2fs; %.2f connections/user sec, bytes read %ld
- %d connections in %ld real seconds, %ld bytes read per connection
- Now timing with session id reuse.
- Unable to get connection
- starting
- apps.c
- %s Policies:
- 
- Require explicit Policy: %s
- Authority
- PKCS12
- Invalid Policy "%s"
- unrecognized purpose
- -verify\_depth
- invalid depth
- -atime
- Error parsing timestamp %s
- -ignore\_critical
- -issuer\_checks
- -crl\_check
- -crlcheckall
- -policy\_check
- -explicit\_policy
- -inhibit\_any
- -inhibit\_map
- -x509\_strict
- -extended\_crl
- -use\_deltas
- -policy\_print
- -checksssig
- Subject does not start with '/'.
- escape character at end of string
- end of string encountered while processing type of subject name element #%d
- Subject Attribute %s has no known NID, skipped
- No value provided for Subject Attribute %s, skipped

- file name too long
- %s.attr
- %s.attr.%s
- unable to rename %s to %s
- reason
- unable to open '%s'
- unique\_subject = %s
- error creating serial number index:(%ld,%ld,%ld)
- error creating name index:(%ld,%ld,%ld)
- error converting serial to ASN.1 format
- Out of memory
- unable to load number from %s
- error converting number from bin to BIGNUM
- Error configuring OpenSSL
- enabling auto ENGINE support
- dynamic
- SO\_PATH
- invalid engine "%s"
- SETUSERINTERFACE
- can't use that engine
- engine "%s" set.
- copyall
- bad input format specified for %s
- Error opening %s %s
- certificates
- unable to load %s
- Error reading %s %s
- no keyfile specified
- no engine specified
- bad input format specified for key file
- Error loading PKCS12 file for %s
- Passphrase callback error for %s
- Mac verify error (wrong password?) in PKCS12 file for %s
- cannot load %s from engine
- Error reading header on certificate
- unable to load certificate
- oid\_section
- problem loading oid section %s
- problem creating object %s=%s
- Can't read environment variable %s
- Can't open file %s
- Can't access file descriptor %s
- Can't open BIO for stdin
- Invalid password argument "%s"
- Error reading password from BIO
- pass phrase
- User interface error
- aborted!
- OpenSSL application user interface
- issuer=
- error on line %ld of db attribute file '%s'
- esc\_2253

- esc\_ctrl
- esc\_msb
- use\_quote
- ignore\_type
- show\_type
- dump\_all
- dump\_nostr
- dump\_der
- compat
- sepcommplus
- sepcommplus\_space
- sepsemplus\_space
- sep\_multiline
- dn\_rev
- nofname
- space\_eq
- dump\_unknown
- RFC2253
- oneline
- multiline
- ca\_default
- compatible
- no\_header
- no\_version
- no\_serial
- no\_signame
- no\_validity
- no\_subject
- no\_issuer
- no\_pubkey
- no\_extensions
- no\_sigdump
- no\_aux
- no\_attributes
- ext\_default
- ext\_error
- ext\_parse
- ext\_dump
- SSL 2.0
- SSL 3.0
- TLS 1.0
- TLS 1.1
- TLS 1.2
- DTLS 1.0
- , CLIENT-CERTIFICATE
- , ERROR:
- UNSUPPORTED-CERTIFICATE-TYPE-ERROR
- NO-CIPHER-ERROR
- BAD-CERTIFICATE-ERROR
- , CLIENT-HELLO
- , CLIENT-MASTER-KEY
- , CLIENT-FINISHED



- , SERVER-HELLO
- , SERVER-VERIFY
- , SERVER-FINISHED
- , REQUEST-CERTIFICATE
- ChangeCipherSpec
- , fatal
- , warning
- close\_notify
- unexpected\_message
- badrecordmac
- decryption\_failed
- record\_overflow
- decompression\_failure
- handshake\_failure
- bad\_certificate
- unsupported\_certificate
- certificate\_revoked
- certificate\_expired
- certificate\_unknown
- illegal\_parameter
- unknown\_ca
- access\_denied
- decode\_error
- decrypt\_error
- export\_restriction
- protocol\_version
- insufficient\_security
- internal\_error
- user\_canceled
- no\_renegotiation
- unsupported\_extension
- certificate\_unobtainable
- unrecognized\_name
- bad*certificate*status\_response
- bad*certificate*hash\_value
- , HelloRequest
- , ClientHello
- , ServerHello
- , HelloVerifyRequest
- , Certificate
- , ServerKeyExchange
- , CertificateRequest
- , ServerHelloDone
- , CertificateVerify
- , ClientKeyExchange
- , HeartbeatRequest
- %s %s%s [length %04lx]%s%s
- unknownpskidentity
- , Finished
- , Heartbeat
- DTLS 1.0 (bad)
- NO-CERTIFICATE-ERROR

- Handshake
- , HeartbeatResponse
- s\_cb.c
- error setting random cookie secret
- TLS padding
- server name
- max fragment length
- client certificate URL
- trusted CA keys
- truncated HMAC
- status request
- user mapping
- client authz
- server authz
- cert type
- elliptic curves
- EC point formats
- signature algorithms
- use SRTP
- heartbeat
- session ticket
- renegotiation info
- next protocol
- server
- client
- TLS %s extension "%s" (id=%d), len=%d
- SSL\_connect
- SSL\_accept
- undefined
- SSL3 alert %s:%s:%s
- %s:failed in %s
- %s:error in %s
- read from %p [%p] (%lu bytes => %ld (0x%lX))
- write to %p [%p] (%lu bytes => %ld (0x%lX))
- error setting certificate
- error setting private key
- Private key does not match the certificate public key
- unable to get certificate from '%s'
- unable to get private key from '%s'
- depth=%d
- 
- verify error:num=%d:%s
- verify return:%d
- getservbyname failure for %s
- %u.%u.%u.%u
- invalid IP address
- gethostbyname failure
- gethostbyname addr is not AF\_INET
- no port defined
- errno=%d
- accept
- bad gethostbyaddr

- s\_socket.c
- socket
- keepalive
- connect
- unable to write 'random state'
- unable to load 'random state'
- This means that the random number generator has not been seeded
- with much random data.
- Consider setting the RANDFILE environment variable to point at a file that
- 'random' data can be kept in (the file will be overwritten).
- usage:version -[avbofpd]
- %s (Library: %s)
- OpenSSL 1.0.1h 5 Jun 2014
- options:
- unable to load SSL\_SESSION
- Context too long
- No certificate present
- unable to write SSL\_SESSION
- unable to write X509
- usage: sess\_id args
- -text - print ssl session id details
- -cert - output certificate
- -context arg - set the session ID context
- Error in cipher list
- 0x%02X,0x%02X,0x%02X -
- 0x%02X,0x%02X -
- 0x%02X,0x%02X,0x%02X,0x%02X -
- usage: ciphers args
- -v - verbose mode, a textual listing of the SSL/TLS ciphers in OpenSSL
- -V - even more verbose
- -ssl2 - SSL2 mode
- -ssl3 - SSL3 mode
- -tls1 - TLS1 mode
- -toseq
- Can't open input file %s
- Error reading certs file %s
- Error reading sequence file %s
- Netscape certificate sequence utility
- Usage nseq [options]
- -in file input file
- -out file output file
- -toseq output NS Sequence file
- %s, Iteration %ld
- Unknown PBE algorithm %s
- %s:
- %s:
- 
- 
- Key bag
- Bag Attributes
- Key Attributes
- Shrouded Keybag:

- Certificate bag
- Safe Contents bag
- Warning unsupported bag type:
- PKCS7 Data
- PKCS7 Encrypted data:
- -nokeys
- -keyex
- -keysig
- -nocerts
- -clcerts
- -cacerts
- -twopass
- -nomacver
- -descert
- -export
- -noiter
- -maciter
- -nomaciter
- -nomac
- -macalg
- -certpbe
- -keypbe
- -caname
- -password
- Usage: pkcs12 [options]
- -export output PKCS12 file
- -chain add certificate chain
- -inkey file private key if not infile
- -certfile f add all certs in f
- -CApath arg - PEM format directory of CA's
- -CAfile arg - PEM format file of CA's
- -name "name" use name as friendly name
- -caname "nm" use nm as CA friendly name (can be used more than once).
- -in infile input filename
- -out outfile output filename
- -noout don't output anything, just verify.
- -nomacver don't verify MAC.
- -nocerts don't output certificates.
- -clcerts only output client certificates.
- -cacerts only output CA certificates.
- -nokeys don't output private keys.
- -info give info about PKCS#12 structure.
- -des encrypt private keys with DES
- -des3 encrypt private keys with triple DES (default)
- -idea encrypt private keys with idea
- -seed encrypt private keys with seed
- -aes128, -aes192, -aes256
- encrypt PEM output with cbc aes
- -camellia128, -camellia192, -camellia256
- encrypt PEM output with cbc camellia
- -nodes don't encrypt private keys
- -noiter don't use encryption iteration

- -nomaciter don't use MAC iteration
- -maciter use MAC iteration
- -nomac don't generate MAC
- -twopass separate MAC, encryption passwords
- -descrypt encrypt PKCS#12 certificates with triple DES (default RC2-40)
- -certpbe alg specify certificate PBE algorithm (default RC2-40)
- -keypbe alg specify private key PBE algorithm (default 3DES)
- -macalg alg digest algorithm used in MAC (default SHA1)
- -keyex set MS key exchange type
- -keysig set MS key signature type
- -password p set import/export password source
- -passin p input file pass phrase source
- -passout p output file pass phrase source
- -engine e use engine e, possibly a hardware device.
- -rand file%cfile%c...
- load the file (or the files in the directory) into
- the random number generator
- -CSP name Microsoft CSP name
- -LMK Add local machine keyset attribute to private key
- 
- Error opening input file %s
- 
- Enter MAC Password:
- Nothing to do!
- private key
- No certificate matches private key
- certificates from certfile
- Memory allocation error
- Error %s getting chain.
- Enter Export Password:
- Unknown digest algorithm %s
- Enter Import Password:
- MAC Iteration %ld
- Mac verify error: invalid password?
- MAC verified OK
- Error outputting keys and certificates
- -topk8
- -nocrypt
- -nooct
- -embed
- Usage pkcs8 [options]
- -inform X input format (DER or PEM)
- -passin arg input file pass phrase source
- -outform X output format (DER or PEM)
- -passout arg output file pass phrase source
- -topk8 output PKCS8 file
- -nooct use (nonstandard) no octet format
- -embed use (nonstandard) embedded DSA parameters format
- -nsdb use (nonstandard) DSA Netscape DB format
- -noiter use 1 as iteration count
- -nocrypt use or expect unencrypted private key
- -v2 alg use PKCS#5 v2.0 and cipher "alg"

- -v1 obj use PKCS#5 v1.5 and cipher "alg"
- Error converting key
- Enter Encryption Password:
- Error encrypting key
- Error reading key
- Enter Password:
- Error decrypting key
- Warning: broken key encoding:
- No Octet String in PrivateKey
- DSA parameters included in PrivateKey
- DSA public key include in PrivateKey
- DSA private key value is negative
- Unknown broken type
- -text\_pub
- Usage pkey [options]
- Usage pkeyparam [options]
- -text print parameters as text
- -noout don't output encoded parameters
- Error reading parameters
- Usage: pkeyutl [options]
- -sigfile file signature file (verify operation only)
- -pubin input is a public key
- -certin input is a certificate carrying a public key
- -pkeyopt X:Y public key options
- -verifyrecover verify with public key, recover original data
- -derive derive shared secret
- -passin arg pass phrase source
- -sigfile
- -peerkey
- -peerkey command before -inkey
- Peer Key
- Error reading peer key %s
- -peerform
- -verifyrecover
- -derive
- -pkeyopt command before -inkey
- Signature file specified for non verify
- No signature file specified for verify
- Error Opening Input File
- Error Creating Output File
- Can't open signature file %s
- Error reading signature data
- Signature Verification Failure
- Signature Verified Successfully
- pkeyutl.c
- Public Key operation error
- -challenge
- -spksect
- %s [options]
- -key arg create SPKAC using private key
- -passin arg input file pass phrase source
- -challenge arg challenge string

- -spkac arg alternative SPKAC name
- -noout don't print SPKAC
- -verify verify SPKAC signature
- -engine e use engine e, possibly a hardware device.
- Error opening output file
- SPKAC=%s
- Error opening input file
- Error parsing config file
- Can't find SPKAC called "%s"
- Error loading SPKAC
- Signature OK
- Signature Failure
- -resign
- -pk7out
- -rc2-40
- -rc2-128
- -rc2-64
- -nointern
- -nochain
- -noattr
- -nodetach
- -nosmimecap
- -nosigs
- -stream
- -indef
- -noindef
- -nooldmime
- -crlfeol
- -signer
- -recip
- Unknown digest %s
- Illegal -inkey without -signer
- -content
- Multiple signers or keys not allowed
- No signer certificate specified
- No recipient certificate or key specified
- No recipient(s) certificate(s) specified
- Usage smime [options] cert.pem ...
- -encrypt encrypt message
- -decrypt decrypt encrypted message
- -sign sign message
- -verify verify signed message
- -pk7out output PKCS#7 structure
- -des3 encrypt with triple DES
- -des encrypt with DES
- -seed encrypt with SEED
- -rc2-40 encrypt with RC2-40 (default)
- -rc2-64 encrypt with RC2-64
- -rc2-128 encrypt with RC2-128
- encrypt PEM output with cbc aes
- encrypt PEM output with cbc camellia
- -nointern don't search certificates in message for signer

- -nosigs don't verify message signature
- -noverify don't verify signers certificate
- -nocerts don't include signers certificate when signing
- -nodetach use opaque signing
- -noattr don't include any signed attributes
- -binary don't translate message to text
- -certfile file other certificates file
- -signer file signer certificate file
- -recip file recipient certificate file for decryption
- -in file input file
- -inform arg input format SMIME (default), PEM or DER
- -inkey file input private key (if not signer or recipient)
- -keyform arg input private key format (PEM or ENGINE)
- -out file output file
- -outform arg output format SMIME (default), PEM or DER
- -content file supply or override content for detached signature
- -to addr to address
- -from ad from address
- -subject s subject
- -text include or delete text MIME headers
- -CApath dir trusted certificates directory
- -CAfile file trusted certificates file
- -crl\_check check revocation status of signer's certificate using CRLs
- -crlcheckall check revocation status of signer's certificate chain using CRLs
- -engine e use engine e, possibly a hardware device.
- -passin arg input file pass phrase source
- load the file (or the files in the directory) into
- cert.pem recipient certificate(s) for encryption
- recipient certificate file
- signing key file
- Bad input format for PKCS#7 file
- Error reading S/MIME message
- Can't read content file %s
- signer certificate
- Error creating PKCS#7 structure
- Error decrypting PKCS#7 structure
- Verification successful
- Verification failure
- To: %s
- From: %s
- Subject: %s
- Bad output format for PKCS#7 file
- Error writing signers to %s
- -sign\_receipt
- -verify\_retcode
- -verify\_receipt
- -cmsout
- -data\_out
- -data\_create
- -digest\_verify
- -digest\_create
- -compress



- -uncompress
- -EncryptedData\_decrypt
- -EncryptedData\_encrypt
- -debug\_decrypt
- -nosignercert\_verify
- -keyid
- -nocontentverify
- -noattrverify
- -receiptrequestprint
- -receiptrequestall
- -receiptrequestfirst
- -receiptrequestfrom
- -receiptrequestto
- -secretkey
- Invalid key %s
- -secretkeyid
- Invalid id %s
- -pwri\_password
- -econtent\_type
- Invalid OID %s
- -certsout
- -rctform
- No Signed Receipts Recipients
- Signed receipts only allowed with -sign
- Usage cms [options] cert.pem ...
- -cmsout output CMS structure
- -certsout file certificate output file
- -keyid use subject key identifier
- No secret key id
- receipt signer certificate file
- Bad input format for CMS file
- Error writing certs to %s
- Can't open receipt file %s
- Bad input format for receipt
- Error reading receipt
- Signed Receipt Request Creation Error
- Error creating CMS structure
- Error decrypting CMS using secret key
- Error decrypting CMS using private key
- Error decrypting CMS using password
- Error decrypting CMS structure
- Signer %d:
- No Receipt Request
- Receipt Request Parse Error
- Signed Content ID:
- Receipts From
- List:
- : First Tier
- Unknown (%d)
- Receipts To:
- Bad output format for CMS file
- Usage: rand [options] num

- -out file - write to file
- -engine e - use engine e, possibly a hardware device.
- -rand file%cfile%c... - seed PRNG from files
- -base64 - base64 encode output
- -hex - hex encode output
- engine.c
- [Error]: internal stack error
- [Error]: command name too long
- [Failure]: %s
- [Success]: %s
- (%s) %s
- Loaded: (%s) %s
- [ available ]
- [ unavailable ]
- 
- %s: %s
- %s%s(input flags):
- 
- [Internal]
- NUMERIC
- STRING
- NO\_INPUT
- <0x%04X>
- 
- usage: engine opts [engine ...]
- -v[v[v[v]]] - verbose mode, for each engine, list its 'control commands'
- -vv will additionally display each command's description
- -vvv will also add the input flags for each command
- -vvvv will also show internal input flags
- -c - for each engine, also list the capabilities
- -t[t] - for each engine, check that they are really available
- -tt will display error trace for unavailable engines
- -pre - runs command 'cmd' against the ENGINE before any attempts to load it (if -t is used)
- -post - runs command 'cmd' against the ENGINE after loading it
- (only used if -t is also provided)
- NB: -pre and -post will be applied to all ENGINES supplied on the command line, or all supported ENGINES if none are specified.
- Eg. '-pre "SOPATH:/lib/libdriver.so"' calls command "SOPATH" with argument "/lib/libdriver.so".
- Error creating connect BIO
- Error creating SSL context.
- Error connecting BIO
- Can't get connection fd
- Timeout on connect
- Unexpected retry condition
- Timeout on request
- Select error
- Error querying OCSP responder
- HTTP/1.0 200 OK
- Content-type: application/ocsp-response
- Content-Length: %d

- Illegal timeout value %s
- -header
- -ignore\_err
- -nonce
- -no\_nonce
- -respnocerts
- -respkeyid
- -no\_certs
- -nosignatureverify
- -nocertverify
- -no\_chain
- -nocertchecks
- -no\_explicit
- -trust\_other
- -no\_intern
- -req\_text
- -resp\_text
- -reqin
- -respin
- -VAfile
- -sign\_other
- -verify\_other
- -validity\_period
- Illegal validity period %s
- -status\_age
- Illegal validity age %s
- -reqout
- -respout
- issuer certificate
- No issuer certificate specified
- Error Creating OCSP request
- Error converting serial number %s
- -index
- Illegal update period %s
- -nrequest
- Illegal accept count %s
- -ndays
- -rsigner
- -rother
- OCSP utility
- Usage ocsf [options]
- -out file output filename
- -issuer file issuer certificate
- -cert file certificate to check
- -serial n serial number to check
- -signer file certificate to sign OCSP request with
- -signkey file private key to sign OCSP request with
- -sign\_other file additional certificates to include in signed request
- -no\_certs don't include any certificates in signed request
- -req\_text print text form of request
- -resp\_text print text form of response
- -text print text form of request and response

- -reqout file write DER encoded OCSP request to "file"
- -respout file write DER encoded OCSP response to "file"
- -reqin file read DER encoded OCSP request from "file"
- -respin file read DER encoded OCSP response from "file"
- -nonce add OCSP nonce to request
- -no\_nonce don't add OCSP nonce to request
- -url URL OCSP responder URL
- -host host:n send OCSP request to host on port n
- -path path to use in OCSP request
- -CApath dir trusted certificates directory
- -CAfile file trusted certificates file
- -VAfile file validator certificates file
- -validity\_period n maximum validity discrepancy in seconds
- -status\_age n maximum status age in seconds
- -noverify don't verify response at all
- -verify\_other file additional certificates to search for signer
- -trust\_other don't verify additional certificates
- -no\_intern don't search certificates contained in response for signer
- -nosignatureverify don't check signature on response
- -nocertverify don't check signing certificate
- -no\_chain don't chain verify response
- -nocertchecks don't do additional checks on signing certificate
- -port num
- port to run responder on
- -index file
- certificate status index file
- -CA file
- CA certificate
- -rsigner file
- responder certificate to sign responses with
- -rkey file
- responder key to sign responses with
- -rother file
- other certificates to include in response
- -respnocerts don't include any certificates in response
- -nmin n
- number of minutes before next update
- -ndays n
- number of days before next update
- -respkeyid identify response by signing certificate key ID
- -nrequest n number of requests to accept (default unlimited)
- - use specified digest in the request
- Error Opening OCSP request file
- Error reading OCSP request
- Error setting up accept BIO
- responder certificate
- Error loading responder certificate
- responder other certificates
- responder private key
- Waiting for OCSP client connections...
- Error accepting connection
- Invalid request

- Error parsing OCSP request
- Need an OCSP request for this operation!
- Error loading signer certificate
- signer certificates
- signer private key
- Error signing OCSP request
- Error opening file %s
- Need a responder certificate, key and CA for this operation!
- ocsd.c
- Error Opening OCSP response file
- Error reading OCSP response
- Responder Error: %s (%d)
- validator certificate
- Error parsing response
- WARNING: no nonce in response
- Nonce Verify error
- Response Verify Failure
- Response verify OK
- ERROR: No Status found.
- WARNING: Status times invalid.
- This Update:
- Next Update:
- Reason: %s
- Revocation Time:
- -generate
- -checks
- Unknown option '%s'
- No prime specified
- Specify the number of bits.
- is %sprime
- %s-14s hex
- %s-14s number of checks
- -checks
- Response is not generated.
- Response has been generated.
- Warning: could not open file %s for reading, using serial number: 1
- Error during serial number generation.
- could not save serial number to %s
- -section
- -query
- -digest
- -token\_in
- -token\_out
- -reply
- -queryfile
- Error getting password.
- bad digest, %d bytes must be specified
- cannot convert %s to OID
- could not create nonce
- could not create query
- invalid digest string
- memory allocation failure

- Verification:
- FAILED
- usage:
- ts -query [-rand file%cfile%c...] [-config configfile] [-data filetohash] [-digest digest\_bytes][-
- ts -reply [-config configfile] [-section tsa\_section] [-queryfile request.tsq] [-passin password] [-
- ts -verify [-data filetohash] [-digest digest\_bytes] [-queryfile request.tsq] -in response.tsr [-t
- %s "%s"
- %d = "%s"
- User entry
- g N entry
- Creating
- user="%s"
- g="%s"
- N="%s"
- Internal error creating SRP verifier
- gNid=%s salt = "%s"
- verifier = "%s"
- -delete
- -modify
- -userinfo
- -dbfile and -configfile cannot be specified together.
- Exactly one of the options -add, -delete, -modify -list must be specified.
- Need at least one user for options -add, -delete, -modify.
- -passin, -passout arguments only valid with one user.
- trying to read defaultsrp in " BASESECTION "
- default\_srp
- trying to read srpvfile in section "%s"
- srpvfile
- Trying to read SRP verifier file "%s"
- Database initialised
- Default g and N
- No g and N value for index "%s"
- Database has no g N information.
- Starting user processing
- Processing user "%s"
- List all users
- user "%s" does not exist, ignored. t
- user "%s" reactivated.
- Cannot create srp verifier for user "%s", operation abandoned .
- failed to update srpvfile
- user "%s" does not exist, operation ignored.
- user "%s" already updated, operation ignored.
- Verifying password for user "%s"
- Validating
- user="%s"
- srp\_verifier="%s"
- srp\_usersalt="%s"
- g="%s"
- N="%s"
- Pass %s
- Internal error validating SRP verifier
- Invalid password for user "%s", operation abandoned.

- Password for user "%s" ok.
- Cannot create srp verifier for user "%s", operation abandoned.
- user "%s" does not exist, operation ignored. t
- user "%s" revoked. t
- User proccession done.
- Trying to update srpvfile.
- Temporary srpvfile created.
- srpvfile updated.
- User errors %d.
- SRP terminating with code %d.
- usage: srp [args] [user]
- -verbose Talk alot while doing things
- -config file A config file
- -name arg The particular srp definition to use
- -srpvfile arg The srp verifier file name
- -add add an user and srp verifier
- -modify modify the srp verifier of an existing user
- -delete delete user from verifier file
- -list list user
- -gn arg g and N values to be used for new verifier
- -userinfo arg additional info to be set for user
- -engine e - use engine e, possibly a hardware device.
- ./0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
- 4ah7=.J[
- q"dYiN
- lz\*!,l
- EwnW`W
- yR
- (w 1q,
- >
- lZL,YV
- 
- e!S\_()O
- (P9gS,<
- "-->Z1
- \*kX\_Hx
- cP"/XXyk
- ^@ H{?
- \_!1trkd
- -time arg - max number of seconds to collect data, default %d
- -verify arg - turn on peer certificate verification, arg == depth
- -cert arg - certificate file to use, PEM format assumed
- -key arg - RSA file to use, PEM format assumed, key is in cert file
- file if not specified by this option
- -CApath arg - PEM format directory of CA's
- -CAfile arg - PEM format file of CA's
- -cipher - preferred cipher to use, play with 'openssl ciphers'
- dyldstubbinding\_helper
- \_dyldfunc\_lookup
- \_SortFnByName
- functionLHASH\_COMP
- functionLHASH\_HASH

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- *docmd*
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- *showciphers*
- *dopasswd*
- *dhcb*
- *dobody*
- *\_certify*
- *writenew\_certificate*
- *dorevoke*
- *dsacb*
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- *genrsacb*
- *genpkeycb*
- *nextproto\_cb*
- *printstats*
- *loaddh\_param*
- *certstatus\_cb*
- *wwwbody*
- *ss/srpserverparam\_cb*
- *ss/servername\_cb*
- *pskserver\_cb*
- *tmprsa\_cb*
- *generatesession\_id*
- *svbody*
- *printstuff*
- *ocspresp\_cb*
- *ss/srpverifyparam\_cb*
- *ss/givesrpclientpwdcb*
- *ss/servername\_cb*
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- *TSRESP*printbio
- *TSRESP*setstatus\_info
- *TSRESP*settst\_info
- *TSRESP*verifyresponse
- *TSRESP*verifytoken
- *TSSTATUSINFO*free
- *TSSTATUSINFO*new
- *TSTSTINFO*free
- *TSTSTINFO*print\_bio
- *TSVERIFYCTX*free
- *TSVERIFYCTX*new
- *TXTDB*createindex
- *TXTDB*\_free
- *TXTDB*getby\_index
- *TXTDB*\_insert
- *TXTDB*\_read
- *TXTDB*\_write
- *UIOpenSSL*
- *Uladdinputstring*
- *Uladdverifystring*

- *Ulconstruct\_prompt*
- *Ulcreate\_method*
- *Ulctrl*
- *Uldestroy\_method*
- *Ulfree*
- *Ulget0userdata*
- *Ulgetinputflags*
- *Ulgetstringtype*
- *UImethodgetcloser*
- *UImethodgetopener*
- *UImethodgetreader*
- *UImethodgetwriter*
- *UImethodsetcloser*
- *UImethodsetopener*
- *UImethodsetreader*
- *UImethodsetwriter*
- *Ulnew\_method*
- *Ulprocess*
- *Ulset\_result*
- *\_WHIRLPOOL*
- *X509V3EXTCRLadd\_nconf*
- *X509V3EXTREQadd\_nconf*
- *X509V3EXTaddnconf*
- *X509V3add\_value*
- *X509V3conf\_free*
- *X509V3parse\_list*
- *X509V3set\_ctx*
- *X509V3set\_nconf*
- *X509ALGOR\_free*
- *X509ALGOR\_new*
- *X509CRLadd0revoked*
- *X509CRLadd1ext\_i2d*
- *X509CRL\_digest*
- *X509CRL\_free*
- *X509CRLgettext\_d2i*
- *X509CRL\_new*
- *X509CRL\_print*
- *X509CRLsetissuer\_name*
- *X509CRLsetlastUpdate*
- *X509CRLsetnextUpdate*
- *X509CRLsetversion*
- *X509CRLsignctx*
- *X509CRL\_sort*
- *X509CRL\_verify*
- *X509EXTENSION\_free*
- *X509EXTENSIONgetobject*
- *X509INFO\_free*
- *X509LOOKUP\_ctrl*
- *X509LOOKUP\_file*
- *X509LOOKUPhashdir*

- *X509NAMEENTRY*free
- *X509NAMEENTRY*get\_data
- *X509NAMEENTRY*get\_object
- *X509NAME*addentry
- *X509NAME*addentrybyNID
- *X509NAME*addentrybytxt
- *X509NAME*deleteentry
- *X509NAME*\_dup
- *X509NAME*entrycount
- *X509NAME*\_free
- *X509NAME*getentry
- *X509NAME*getindexbyNID
- *X509NAME*getindexbyOBJ
- *X509NAME*\_hash
- *X509NAME*hashold
- *X509NAME*\_new
- *X509NAME*\_online
- *X509NAME*printex
- *X509NAME*printex\_fp
- *X509OBJECT*freecontents
- *X509POLICYNODE*print
- *X509PURPOSE*\_get0
- *X509PURPOSE*get0name
- *X509PURPOSE*get0sname
- *X509PURPOSE*getby\_sname
- *X509PURPOSE*getcount
- *X509PURPOSE*getid
- *X509REQ*add1attrbyNID
- *X509REQ*add1attrbytxt
- *X509REQ*checkprivate\_key
- *X509REQ*\_free
- *X509REQ*getextensions
- *X509REQ*getpubkey
- *X509REQ*\_new
- *X509REQ*\_print
- *X509REQ*printex
- *X509REQ*setpubkey
- *X509REQ*setsubject\_name
- *X509REQ*setversion
- *X509REQ*signctx
- *X509REQ*\_verify
- *X509REVOKED*add1ext\_i2d
- *X509REVOKED*\_new
- *X509REVOKED*setrevocationDate
- *X509REVOKED*setserialNumber
- *X509SIG*\_free
- *X509STORE*CTXcleanup
- *X509STORE*CTXfree
- *X509STORE*CTXget0parentctx
- *X509STORE*CTXget0policytree



- X509STORECTXget1\_chain
- X509STORECTXgetcurrentcert
- X509STORECTXget\_error
- X509STORECTXgeterrordepth
- X509STORECTXgetexplicitpolicy
- X509STORECTXinit
- X509STORECTXnew
- X509STORECTXset0\_crls
- X509STORECTXset\_cert
- X509STORECTXset\_flags
- X509STORECTXtrusted\_stack
- X509STOREaddlookup
- X509STORE\_free
- X509STOREgetby\_subject
- X509STOREloadlocations
- X509STORE\_new
- X509STOREset1param
- X509STOREsetdefault\_paths
- X509STOREsetflags
- X509STOREsetverify\_cb
- X509VERIFYPARAMadd0\_policy
- X509VERIFYPARAMfree
- X509VERIFYPARAMnew
- X509VERIFYPARAMset\_depth
- X509VERIFYPARAMset\_flags
- X509VERIFYPARAMset\_purpose
- X509VERIFYPARAMset\_time
- X509add1rejectobject
- X509add1trustobject
- X509add\_ext
- X509alias\_get0
- X509alias\_set1
- X509checkprivatekey
- X509check\_purpose
- X509cmp\_time
- X509delete\_ext
- X509digest
- X509email\_free
- X509free
- X509get0pubkeybitstr
- X509get1\_email
- X509get1\_ocsp
- X509getdefaultcert\_area
- X509get\_ext
- X509getextby\_OBJ
- X509gettextcount
- X509getissuename
- X509get\_pubkey
- X509get\_serialNumber
- X509getsubjectname

- X509gmtime\_adj
- X509issuenamehash
- X509issuenamehash\_old
- X509keyid\_set1
- X509new
- X509ocspid\_print
- X509policytreeget0\_policies
- X509policytreeget0userpolicies
- X509print
- X509print\_ex
- X509reject\_clear
- X509set issuename
- X509set\_pubkey
- X509set\_serialNumber
- X509setsubjectname
- X509set\_version
- X509sign
- X509sign\_ctx
- X509subjectnamehash
- X509subjectnamehash\_old
- X509timeadjex
- X509toX509REQ
- X509trust\_clear
- X509verify
- X509verify\_cert
- X509verifycerterror\_string
- \_\_DefaultRuneLocale
- \_assert\_rtn
- \_error
- \_maskrune
- \_stderrp
- \_stdinp
- \_stdoutp
- \_toupper
- a2iASN1\_INTEGER
- a2iGENERAL\_NAME
- \_abort
- \_accept\$UNIX2003
- \_access
- \_alarm
- b2iPVK\_bio
- b2iPrivateKey\_bio
- b2iPublicKey\_bio
- \_bind\$UNIX2003
- \_close\$UNIX2003
- \_connect\$UNIX2003
- d2iASN1\_TYPE
- d2iCMS\_bio
- d2iDHparams
- d2iDSAparams

- *d2iECPKParameters*
- *d2iECPrivateKey\_bio*
- *d2iECPUBKEYbio*
- *d2iOCSP\_REQUEST*
- *d2iOCSP\_RESPONSE*
- *d2iPBEPARAM*
- *d2iPKCS12\_bio*
- *d2iPKCS7\_bio*
- *d2iPKCS8PRIVKEYINFObio*
- *d2iPKCS8\_bio*
- *d2iPUBKEY\_bio*
- *d2iPrivateKey\_bio*
- *d2iRSAPrivateKey*
- *d2iRSAPublicKey\_bio*
- *d2iRSA\_NET*
- *d2iSSL\_SESSION*
- *d2iTSREQbio*
- *d2iTSRESPbio*
- *d2iX509CRLbio*
- *d2iX509REQbio*
- *d2iX509\_bio*
- *\_fclose*
- *\_fdopen*
- *\_fflush*
- *\_fgetc*
- *\_fileno*
- *\_fopen*
- *\_fprintf\$LDBL128*
- *\_fputc*
- *\_fputs\$UNIX2003*
- *\_fwrite\$UNIX2003*
- *\_getenv*
- *\_gethostbyaddr*
- *\_gethostbyname*
- *\_getservbyname*
- *\_getsockname\$UNIX2003*
- *i2aASN1\_INTEGER*
- *i2aASN1\_OBJECT*
- *i2bPVK\_bio*
- *i2bPrivateKey\_bio*
- *i2bPublicKey\_bio*
- *i2dASN1\_TYPE*
- *i2dCMSbiostream*
- *i2dDHparams*
- *i2dDSAPrivateKey\_bio*
- *i2dDSAPUBKEYbio*
- *i2dDSAPparams*
- *i2dECPKParameters*
- *i2dECPrivateKey\_bio*
- *i2dECPUBKEYbio*
- *i2dOCSP\_REQUEST*

- *i2dOCSP\_RESPONSE*
- *i2dPKCS12\_bio*
- *i2dPKCS7\_bio*
- *i2dPKCS7biostream*
- *i2dPKCS8PRIVKEYINFObio*
- *i2dPKCS8\_bio*
- *i2dPUBKEY\_bio*
- *i2dPrivateKey\_bio*
- *i2dRSAPrivateKey\_bio*
- *i2dRSAPublicKey\_bio*
- *i2dRSA\_NET*
- *i2dRSAPUBKEYbio*
- *i2dSSL\_SESSION*
- *i2dTSREQbio*
- *i2dTSRESPbio*
- *i2dX509*
- *i2dX509CRLbio*
- *i2dX509\_NAME*
- *i2dX509\_PUBKEY*
- *i2dX509REQbio*
- *i2dX509\_bio*
- *ideacbc\_encrypt*
- *ideaoptions*
- *ideasetencryptkey*
- *lhfree*
- *lhinsert*
- *lhnew*
- *lhnodestatsbio*
- *lhnodeusagestats\_bio*
- *lhretrieve*
- *lhstats\_bio*
- *lhstrhash*
- *\_listen\$UNIX2003*
- *\_malloc*
- *\_memcmp*
- *\_memcpy*
- *\_memset*
- *\_perror*
- *\_printf\$LDBL128*
- *\_putchar*
- *\_qsort*
- *\_read\$UNIX2003*
- *\_realloc*
- *\_rename*
- *s2iASN1\_INTEGER*
- *\_select\$UNIX2003*
- *\_setsockopt*
- *\_setvbuf*
- *\_shutdown*
- *\_signal*
- *skdelete*

- *skfree*
- *sknew\_null*
- *sknum*
- *skpop\_free*
- *skpush*
- *skshift*
- *skvalue*
- *\_sleep\$UNIX2003*
- *\_socket*
- *\_sscanf\$LDBL128*
- *\_strcasecmp*
- *\_strchr*
- *\_strcmp*
- *\_strcpy*
- *stringto\_hex*
- *\_strlen*
- *\_strncat*
- *\_strncmp*
- *\_strchr*
- *\_strspn*
- *\_strstr*
- *\_sysconf*
- *\_times*
- *\_write\$UNIX2003*

## UTF-8 Strings

No detected.

## UTF-16LE Strings

No detected.

## Extracted URLs

- <http://etherx.jabber.org/streams>

## Base64 Strings

- Possible Base64 Encoded String: blowfish -> Decoded: nZ0~+!
- Possible Base64 Encoded String: blowfish -> Decoded: nZ0~+!

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## File Entropy Analysis

- **Entropy:** 6.24
- **Status:** This file is likely packed or encrypted.

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## VirusTotal Analysis

**Scan Date:** 2021-08-08 19:34:23

## Scan Results

- Malicious: 0
- Suspicious: 0
- Undetected: 61
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 1
- Failure: 1
- Type-unsupported: 13

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

| Engine Name | Detection Category | Detection  |
|-------------|--------------------|------------|
| No results  | No results         | No results |

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